

# Apollo Diagnostics - Laboratory Investigation Report

123, Nungambakkam High Road, Chennai - 600034 | NABL Accredited | Tel: 044-2827-5000

|                  |                 |              |                                        |
|------------------|-----------------|--------------|----------------------------------------|
| Patient Name:    | Saroja Devi     | Report ID:   | APL-MULT-59-SAR-2026                   |
| Age / Gender:    | 59 Yrs / Female | Category:    | Multi-condition: Hypothyroid + Anaemia |
| Weight / Height: | 80 kg / 158 cm  | BMI:         | 32.0 kg/m2                             |
| Ref. Doctor:     | Dr. Arun Sharma | Report Date: | 05-Jan-2026                            |

## 1. COMPLETE BLOOD COUNT [Circulatory System]

### Red Cell Indices

| Test                     | Result | Unit | Reference Range | Flag |
|--------------------------|--------|------|-----------------|------|
| Haemoglobin (Hb)         | 9.5    | g/dL | 12.0 - 16.0     | LOW  |
| Packed Cell Volume (PCV) | 27.6   | %    | 36 - 46         | LOW  |
| MCV                      | 78.2   | fL   | 83 - 101        | LOW  |
| MCH                      | 25.0   | pg   | 27 - 32         | LOW  |
| RDW-CV                   | 16.3   | %    | 11.6 - 14.0     | HIGH |

**Haematology Comment:** Haemoglobin 9.5 g/dL - microcytic (MCV 78.2 fL) hypochromic (MCH 25.0 pg) anaemia pattern. Raised RDW-CV (16.3%) indicates anisocytosis. Consistent with iron deficiency or thalassaemia trait. Further iron studies recommended.

### White Cells & Platelets

| Test           | Result | Unit      | Reference Range | Flag   |
|----------------|--------|-----------|-----------------|--------|
| Total WBC      | 7000   | cells/cmm | 4500 - 11000    | Normal |
| Neutrophils    | 62     | %         | 40 - 75         | Normal |
| Lymphocytes    | 28     | %         | 20 - 45         | Normal |
| Platelet Count | 2.8    | lakh/cmm  | 1.5 - 4.0       | Normal |

**Comment:** WBC and platelet counts within reference limits. No significant leucocytosis, leucopenia or thrombocytopenia.

## 2. IRON STUDIES [Haematopoietic System]

| Test                   | Result | Unit  | Reference Range | Flag |
|------------------------|--------|-------|-----------------|------|
| Serum Iron             | 52     | ug/dL | 60 - 170        | LOW  |
| TIBC                   | 395    | ug/dL | 250 - 370       | HIGH |
| Transferrin Saturation | 10.9   | %     | 20 - 50         | LOW  |
| Serum Ferritin         | 6.7    | ng/mL | 12 - 150        | LOW  |

**Iron Studies - Critical Findings:** Serum iron 52 ug/dL and ferritin 7 ng/mL are reduced. Transferrin saturation 10.9% (normal >20%).  
**RECOMMENDATION:** Iron supplementation 60 mg elemental iron/day. Investigate dietary adequacy and occult GI blood loss.

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## 3. HEPATIC FUNCTION TEST [Hepatobiliary System]

### Liver Enzymes

| Test                       | Result | Unit  | Reference Range | Flag   |
|----------------------------|--------|-------|-----------------|--------|
| SGPT / ALT                 | 22     | U/L   | 7 - 56          | Normal |
| SGOT / AST                 | 20     | U/L   | 10 - 40         | Normal |
| Alkaline Phosphatase (ALP) | 70     | U/L   | 44 - 147        | Normal |
| Total Bilirubin            | 0.6    | mg/dL | 0.2 - 1.2       | Normal |
| Albumin                    | 4.2    | g/dL  | 3.5 - 5.0       | Normal |

Hepatologist Note: Liver enzymes within normal limits. No hepatocellular damage or cholestatic process detected.

## 4. RENAL FUNCTION TEST [Urinary System]

| Test                | Result | Unit          | Reference Range | Flag   |
|---------------------|--------|---------------|-----------------|--------|
| Serum Creatinine    | 1.0    | mg/dL         | 0.6 - 1.1       | Normal |
| Blood Urea Nitrogen | 22     | mg/dL         | 7 - 20          | HIGH   |
| eGFR (CKD-EPI)      | 98     | mL/min/1.73m2 | 60 - 120        | Normal |
| Serum Uric Acid     | 4.8    | mg/dL         | 2.4 - 6.0       | Normal |
| Sodium (Na+)        | 139    | mEq/L         | 135 - 145       | Normal |
| Potassium (K+)      | 4.0    | mEq/L         | 3.5 - 5.1       | Normal |

Nephrologist Note: eGFR 98 mL/min/1.73m2 - glomerular filtration well preserved. Renal function normal.

## 5. LIPID PROFILE [Cardiovascular Risk]

| Test              | Result | Unit  | Reference Range | Flag |
|-------------------|--------|-------|-----------------|------|
| Total Cholesterol | 250    | mg/dL | 0 - 200         | HIGH |
| HDL Cholesterol   | 47     | mg/dL | 50 - 999        | LOW  |
| LDL Cholesterol   | 181    | mg/dL | 0 - 100         | HIGH |
| Triglycerides     | 195    | mg/dL | 0 - 150         | HIGH |

Cardiologist Note: LDL 181 mg/dL (target <100 mg/dL) and Total Cholesterol 250 mg/dL are elevated. Triglycerides 195 mg/dL. Increased cardiovascular risk. RECOMMENDATION: Mediterranean diet, aerobic exercise, statin therapy consideration if >10-year CV risk >7.5%.

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## 6. THYROID FUNCTION TEST [Endocrine System]

| Test    | Result | Unit   | Reference Range | Flag |
|---------|--------|--------|-----------------|------|
| TSH     | 30.3   | uIU/mL | 0.4 - 4.0       | HIGH |
| Free T3 | 1.9    | pg/mL  | 2.3 - 4.2       | LOW  |
| Free T4 | 0.8    | ng/dL  | 0.8 - 1.8       | LOW  |

**Endocrinologist Comment:** TSH: 30.3 uIU/mL is elevated - consistent with hypothyroidism. FT4 0.8 ng/dL is low (overt hypothyroidism). Thyroid antibodies (Anti-TPO) and ultrasound recommended. Consider levothyroxine therapy.

## 7. DIABETES SCREENING [Endocrine - Pancreas]

| Test                        | Result | Unit  | Reference Range | Flag   |
|-----------------------------|--------|-------|-----------------|--------|
| Fasting Blood Glucose       | 88     | mg/dL | 70 - 100        | Normal |
| Post-Prandial Glucose (2hr) | 115    | mg/dL | 0 - 140         | Normal |
| HbA1c                       | 5.1    | %     | 0 - 5.7         | Normal |
| HOMA-IR                     | 1.5    |       | 0 - 2.5         | Normal |

**Endocrinology Note:** Glucose metabolism normal. HbA1c 5.1% - no diabetes or pre-diabetes. HOMA-IR 1.5 - no insulin resistance.

## 8. VITAMINS & BONE [Nutritional / Skeletal]

| Test              | Result | Unit  | Reference Range | Flag   |
|-------------------|--------|-------|-----------------|--------|
| Vitamin D (25-OH) | 16.8   | ng/mL | 30 - 100        | LOW    |
| Vitamin B12       | 450    | pg/mL | 211 - 911       | Normal |
| Folate (Serum)    | 12.0   | ng/mL | 5.4 - 999       | Normal |
| Calcium (Total)   | 8.5    | mg/dL | 8.5 - 10.5      | Normal |
| PTH               | 32     | pg/mL | 15 - 65         | Normal |

**Nutritionist/Endocrinologist Comment:** Vitamin D: 17 ng/mL - deficient (10-20). PTH at 32 pg/mL is within range. Supplementation: 60,000 IU cholecalciferol weekly x 8 weeks, then 1000-2000 IU/day. Recheck in 3 months.

## 9. INFLAMMATORY MARKERS [Immune System]

| Test   | Result | Unit  | Reference Range | Flag   |
|--------|--------|-------|-----------------|--------|
| ESR    | 28     | mm/hr | 0 - 20          | HIGH   |
| CRP    | 5.2    | mg/L  | 0 - 5           | HIGH   |
| hs-CRP | 0.6    | mg/L  | 0 - 1.0         | Normal |

**Immunology Comment:** CRP 5.2 mg/L and ESR 28 mm/hr are elevated. hs-CRP 0.6 mg/L - low. Inflammatory markers elevated - correlate with clinical symptoms and autoimmune screen.

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## CLINICAL SUMMARY & RECOMMENDATIONS

|             |                                                                                                      |
|-------------|------------------------------------------------------------------------------------------------------|
| [ABNORMAL]  | Anaemia: Hb=9.5 g/dL (LOW). Iron deficiency pattern.                                                 |
| [CRITICAL]  | Iron Deficiency: Serum Iron=52 ug/dL, Ferritin=7 ng/mL. Begin iron supplementation.                  |
| [ELEVATED]  | LDL=181 mg/dL (HIGH). Total Cholesterol=250. Dyslipidaemia - dietary + pharmacological management.   |
| [ABNORMAL]  | Hypothyroidism: TSH=30.3 uIU/mL. Levothyroxine therapy evaluation required.                          |
| [DEFICIENT] | Vitamin D=17 ng/mL (LOW). Supplementation required.                                                  |
| [HIGH]      | CRP=5.2 mg/L, hs-CRP=0.6 mg/L - systemic inflammation present.                                       |
| [ACTION]    | Next appointment: Dr. Arun Sharma in 4-8 weeks for follow-up and review of supplementation response. |

Report reviewed by: Dr. Anitha Krishnamurthy, MD (Pathology) | Verified by Dr. Arun Sharma

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